



Aviation Safety Analysis

DATA-DRIVEN INSIGHTS



Overview

- Aviation plays a critical role in transportation sector. This presentation analyzes aviation accident data to identify safety trends and support informed decision making.



Target Audience

- Head of Aviation Division
- Non-technical business leader

Business Problem

- The company does not have aviation experience on the safety side of things.
- Poor aircraft selection might result in financial losses because it will involve a lot of maintenance and high insurance cost.
- It can lead to reputational damage



Our method: The process

- Historical aviation accident data from 1962-2023 analyzed
- Focused on:
 - -Aircraft manufacturers
 - -Aircraft models
 - -Injury severity and fatalities
 - -Used descriptive analytics and clear visualizations to assess risk

Findings

- Some aircraft manufacturers have kept fatality counts per year with a great consistency
- .Certain aircraft models involved in accidents at a higher rate
- Most accidents involves injuries only so selection can occur based on risk

Results

- Provides data driven results that would guide aircraft acquisition.
- It reduces safety, financial and operational risks.
Lays the ground for future aviation analytics